Sahaj Bhadja

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Summary

Backend and platform engineer with 2+ years building **Python services, event-driven pipelines, relational data systems, and cloud infrastructure** with strong ownership of correctness, reliability, observability, and production operations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Advanced NLP, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built an Azure-backed multi-tenant platform with **normalized schemas, row-level isolation, and indexed queries**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily campus users.
- Reworked asynchronous FastAPI orchestration with **idempotent jobs, state transitions, retry boundaries, failure isolation, metrics, and logs**, tripling throughput for 1K+ concurrent AI workloads across stable production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask workflows backed by **REST APIs and relational data models**, supporting 150+ instructors and 10K+ daily production requests for uploads, grading history, and feedback.
- Implemented concurrent Python ingestion with **durable processing, validation, error classification, and operational debugging** for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing peak feedback turnaround 60% daily.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and AWS data pipelines using **object-oriented service boundaries, Glue, S3, and SQL**, improving response times 30% while reducing technical debt 20% across critical integrations.
- Automated 20+ unit and integration suites using PyTest, Mockito, Jenkins, and virtual machines, strengthening **schema compatibility, release validation, production debugging, and cross-platform reliability** across distributed customer services.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, and permissions, synchronizing 2M+ records through **REST APIs, WebSockets, relational schemas, authorization controls, and event-driven integrations** across distributed locations.
- Automated Docker and Jenkins releases, reducing deployment time 50% while applying **Sentry monitoring, structured logging, incident diagnosis, and rollback-ready deployment practices** across eight customer-facing production service environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client workflows into technical specifications for 12+ integrations, mapping bookings, payments, coupons, and statuses into **reliable Flask APIs, Firebase schemas, webhooks, and auditable production data flows**.
- Built responsive interfaces and payment services processing 50K+ records with sub-200-millisecond responses, adding **input validation, authentication, failure handling, and integration testing** for dependable customer transactions across platforms.

Skills

Backend: Python, FastAPI, Flask, Java, Node.js, REST APIs, gRPC, AsyncIO, Microservices, Pydantic, Webhooks, API Design
Data & Streaming: PostgreSQL, MySQL, SQL, Kafka, Debezium, Change Data Capture, Temporal, Schema Evolution, Data Modeling, Event-Driven Systems, Dead-Letter Queues
Correctness & Identity: Idempotency, Event Ordering, Retry Policies, Failure Classification, Service Accounts, API Keys, Entitlements, RBAC, OAuth 2.0, JWT, Least Privilege
Cloud & Reliability: AWS, Azure, GCP, ECS, Fargate, S3, RDS, Terraform, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, OpenTelemetry, Metrics, Traces, Logs, SLOs, Incident Response
Engineering: Git, Linux, Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Code Reviews, Agile, Technical Documentation, On-Call Operations

Projects

IdentityGraph Backbone | *Python, FastAPI, PostgreSQL, Kafka, AWS, gRPC, Terraform*

- Designed an **NHI ingestion backbone** normalizing service accounts, API keys, workload identities, owners, and entitlements from third-party connectors through versioned schemas, idempotent Kafka consumers, ordering keys, and gRPC APIs.

SyncGuard Control Plane | *Python, Temporal, Debezium, Kafka, OpenTelemetry, AWS*

- Built a **reliable synchronization control plane** separating transient retries from permanent failures, routing poison events to dead-letter queues, tracking freshness SLOs, and correlating metrics, traces, and structured logs.

Linguistix | *Python, AWS ECS, Fargate, Docker, CodePipeline, CI/CD*

- Deployed a **containerized Python language service** on AWS ECS and Fargate, automating image builds and releases through CodePipeline while documenting repeatable operations, deployment checks, and production troubleshooting workflows.